



Zihao-Eric GUO

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📍 Châtenay-Malabry, France

🏆 Kaggle Expert

🏠 [Welcome to my homepage](#)

EDUCATION

• Institut Polytechnique de Paris (IPP)

Since 11/2023 – Paris, France

Preserving graph performance in an exogenously perturbed environment - Expected graduation date: **Dec.2026**

• École Centrale de Nantes (ECN)

09/2021 – 09/2023 – Nantes, France

Generalist Engineer Degree (MSc)
- Applied Mathematics Options

• Université de Picardie Jules Verne (UPJV)

09/2017 – 07/2021 – Amiens, France

Bachelor of Science (BSc), Optimization

IT SKILLS

Python, SQL, R, PyTorch,
Power BI, Bash, C/C++, NoSQL,
Gurobi/CPLEX

LANGUAGES

• **Chinese:** Native language

• **English:** Fluent (GRE 326)

• **French:** Fluent (DALF C1)

INTERESTS



STRENGTHS

- Teamwork;
- Autonomy and adaptability;
- Strong learning ability.

📄 Pre-Doc | Seeking a first permanent position

Focus: optimization, machine learning, generative AI



Data Scientist | AI Engineer | Algorithm Engineer



INDUSTRIAL EXPERIENCE

2025

Shanghai,
CHINA

Internship – Algorithm Engineer (Huawei)

08/2025 – 11/2025

- Developed a predictive model based on multimodal contrastive learning, combining two data modalities: 3D point clouds and channel data.
- Designed a non-contiguous resource-allocation optimization algorithm and evaluated it against the RIV reference approach.

2023

Cambridge,
UK

Internship – Biostatistics Researcher Intern (EMBL-EBI)

04/2023 – 09/2023

- Contributed to a maximum-likelihood phylogenetic algorithm for up to 500,000 genomes and developed an automated Python/R/Bash simulation and cluster-benchmarking pipeline.
- Benchmarked topological accuracy, likelihood, runtime and memory over 10 replicates per configuration, demonstrating a speedup of over 100x versus the reference method.

2022

Paris,
France

Internship – Data Engineer (IPEN)

04/2022 – 08/2022

- Analyzed public-transport validations and built a KPI dashboard.
- Compared MATSim with observations to identify discrepancies.
- Applied SARIMAX and NeuralProphet forecasting to analyze drivers and correct statistics.

2021

Nantes,
France
remote

Project - Data Analyst (Tencent)

11/2021 – 01/2022

- Built a scikit-learn churn classifier to support retention and win-back strategies.
- Evaluated it with ROC-AUC and tuned hyperparameters using GridSearchCV.



SCIENTIFIC RESEARCH EXPERIENCE

Selected Publications

(For more publications, please visit my [Google Scholar profile](#).)

CNN text classification method based on simulated annealing method (ACM International Conference Proceeding Series)

Rate variation and recurrent sequence errors in pandemic-scale phylogenetics (Nature Methods)

Multi-task learning for Bayesian Networks

Oct. 2022 – Apr. 2023

Nantes, France – Nantes Digital Science Laboratory (LS2N)

Keywords: C++, Bayesian Networks, Transfer learning

LLM-related Research

Since 2025

- Developed **EvoHGS**, an LLM framework for Hybrid Genetic Search on VRPs; its fine-tuned 8B model outperformed GPT-5 nano (estimated 2B parameters) and achieved SOTA results.
- Adapted the MiniMind pipeline to experiment with an end-to-end lightweight LLM workflow.

Keywords: C++, Reinforcement fine-tuning (RFT), Supervised Fine-Tuning, LoRA, GRPO